Towards Certifiable Data Selection: Statistical Limits and Scalable Algorithms

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Abstract

What does it mean to have an "optimal" data selection algorithm? Optimality is conventionally framed as finding a subset that minimizes the population error of the classifier trained only on that subset. In this work, we show that this goal is information-theoretically impossible: for any compression fraction $\lambda \in (0, 1)$, even when an accurate subset exists in the training set, identifying it requires a validation set strictly larger than the training set itself. This impossibility result motivates shifting from unverifiable population risk minimization to a certifiable, one-sided guarantee that we refer to as *faithfulness*: for any candidate subset, we provide a computable certificate that upper-bounds the difference between training on the full training data and on only a subset. By modeling training dynamics via the Unadjusted Langevin Algorithm, we derive a closed-form upper bound on the Kullback–Leibler divergence between the parameter distributions induced by subset and full-data training in terms of trajectory-averaged gradient discrepancy. This naturally motivates a principled way to obtain a subset by minimizing the trajectory-averaged gradient discrepancy. Even though the resulting problem is NP-hard, we provide an approximation algorithm that is both computationally efficient and equipped with a sub-optimality certificate. Finally, we provide experimental results on various benchmarks showing that our approach outperforms existing data selection baselines.

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1 INTRODUCTION

Training on ever-larger datasets is expensive, and not all samples contribute equally: careful pruning can improve how performance scales with dataset size (Sorscher et al., 2023). *Data selection* formalizes this idea: given a training set S of n samples and a learning algorithm $\mathcal{A}$, find a subset $I \subseteq S$ of size $m < n$ such that training on I alone performs as well as training on all of S. Besides reducing computation and storage, selection is a concrete instance of the data-centric view of machine learning (Zha et al., 2025), in which the dataset, rather than the model, is the object to curate. Yet what makes a subset *good*, and what a selector can verify about it from the training data or validation data is seldom made precise.

Several approaches have been proposed to perform data selection, and have been designed to tackle specific use cases. Bi-level optimization approaches (Killamsetty et al., 2021b; Xia et al., 2024) select subsets of the training set based on the error on a held-out validation set, and training-data based method use only information stemming from the training set and the learning dynamics induced by a given learning algorithm (Toneva et al., 2019; Paul et al., 2021; Killamsetty et al., 2021a). Yet, the statistical validity of these optimization objectives is less studied.

Many data selection methods have been proposed, and they differ mainly in the information they use. Validation-based (bilevel) methods choose the subset to minimize the error on held-out data (Mirzasoleiman et al., 2020; Xia et al., 2024), and therefore target the population risk directly. Training-based methods use the training set and the learning dynamics induced by the learning algorithm: they score examples by forgetting events or gradient norms (Toneva et al., 2019; Paul et al., 2021), or select subsets whose gradient matches that of the full data (Killamsetty et al., 2021a), and thereby implicitly try to reproduce full-data training. Each of these objectives is a proxy, and little is known about what it guarantees; recent theoretical work indicates that selecting from samples is statistically subtle.

Kolossov et al. (2023) show settings where popular selection methods fail, and that the best selection methods vary according to the selection budget. Hanneke et al. (2025a) compare data selection to empirical risk minimizers on the whole training set, and provide settings in which matching ERM performance is impossible.

This leaves open many questions of what a data selector can target and verify about the subset it returns. In this work, we answer the following questions: **(Q1)** *What is achievable from samples in data selection?* **(Q2)** *If risk-optimal selection is out of reach, what can we certify and compute at scale from the training data alone?*

Contributions.

- **Answer to Q1: Sometimes, we need many more validation samples than training samples.** The excess risk of a selected subset can be written in two ways: as an *approximation error* (the best any size-m subset can do) plus a *selection error* (the gap to that best subset) Equation (2), or as the excess risk of full-data training plus a *faithfulness gap* (what is lost by training on the subset instead of on all of S) Equation (6). The first decomposition depends on the unknown distribution P, while the second is available to the selector. This distinguishes validation-based (bilevel) methods, which target the true risk directly, from training-based methods, which try to reproduce full-data training. In Section 3 we study the first view and show that there exist settings where an accurate subset exists, yet no selector can find it unless it either has oracle access to P or a validation set much larger than the training set (Theorem 3.1). This is a statistical limit on what can be identified from samples.

- **Answer to Q2: faithfulness, certified by a training-only bound.** The negative result points to the second decomposition: we ask for a *faithful* subset, one whose training approximates the same distribution over models as training on S. By modeling training dynamics with the Unadjusted Langevin Algorithm (ULA), we bound the faithfulness by the trajectory-averaged gradient discrepancy $G_T(I)$ (Theorem 5.1). This gives a *faithfulness certificate* for any candidate subset. In Figure 1 we show that $G_T(I)$ correlates with subset quality under standard optimizers.

- **A simple algorithm with a suboptimality certificate.** Minimizing $G_T(I)$ is an NP-hard problem, and we do not claim to solve it optimally. Instead, we give a simple swap-descent algorithm that comes with a computable suboptimality certificate (Theorem 5.5). On CIFAR-10 and CIFAR-100, this simple method is on par with existing static data selection baselines. This opens avenues for improving optimization of $G_T(I)$ as a way to improve data selection methods.

2 PRELIMINARIES

2.1 Notation

Let $\mathcal{Z}$ denote the space of data points and $P \in \mathcal{P}(\mathcal{Z})$ denote the underlying data distribution. We consider the setting of empirical risk minimization (ERM) on *i.i.d.* training samples $S = \{Z_1, \ldots, Z_n\} \sim P^n$. A data selection procedure with budget m identifies a subset of the training set $I \subseteq S$ with cardinality $|I| \leq m$. Models are parameterized by a vector $\theta \in \mathbb{R}^d$, and the loss incurred on an individual sample z_i is given by $\ell(\theta, z_i)$. The full-dataset empirical risk is defined as $L_S(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(\theta, z_i)$, while the corresponding subset empirical risk is denoted by $L_I(\theta) = \frac{1}{|I|}\sum_{z \in I} l(\theta, z)$. For a given (randomized) learning algorithm $\mathcal{A} : S \to \mathbb{R}^d$, the population risk of a parameterized hypothesis $\theta \sim \mathcal{A}(I)$ trained on subset I is defined as $R(I) := \mathbb{E}_{Z\sim P, \theta\sim\mathcal{A}(I)}[\ell(\theta, Z)]$, while the optimal population risk over the entire hypothesis class is given by $F^* := \inf_\theta \mathbb{E}_{Z\sim P}[\ell(\theta, Z)]$.

To capture the optimization dynamics of randomized learning procedures, we let $\mathcal{A}_k(S)$ and $\mathcal{A}_k(I)$ denote the time-dependent marginal probability distributions of the model parameters $\theta_k \in \mathbb{R}^d$ at iteration k when trained on the full dataset S and the subset I, respectively. At the final training iteration T, these endpoint parameter distributions are denoted by $\mathcal{A}_T(S)$ and $\mathcal{A}_T(I)$. We measure the discrepancy between two probability distributions P, Q by the Kullback-Leibler divergence (KL): $D_{\mathrm{KL}}(P\|Q) := \mathbb{E}_{x\sim P}[\log(P(x)/Q(x))]$.

2.2 Related Work

MH:discuss this work as well Hanneke et al. (2025b); Coleman et al. (2019)

Data selection methods differ less in how they choose examples than in what they implicitly try to preserve. Methods like G

Data selection methods can be grouped by the quantity their selection criterion is built around. Validation- and target-based methods optimize performance on held-out or target data: Glister maximizes the validation log-likelihood (Killamsetty et al., 2021b), DsDm (Engstrom et al., 2024)chooses the subset that minimizes the trained model's loss on given target tasks, and LESS selects instruction-tuning examples whose

low-dimensional gradient features resemble those of a few target examples (Xia et al., 2024). Other methods build the criterion from the training data: CRAIG and Grad-Match selects a weighted subset whose gradient approximates the full-data gradient (Mirzasoleiman et al., 2020; Killamsetty et al., 2021a), forgetting events (Toneva et al., 2019) and gradient norms (Paul et al., 2021) score examples so that unimportant ones can be pruned. Coverage-based methods work in a feature space: herding picks points whose feature mean tracks the full-data mean (Welling, 2009), facility location and graph cut maximize submodular representation and diversity objectives (Wei et al., 2015; Iyer et al., 2021), and Moderate Coreset keeps points whose distance to their class center is close to the median (Xia et al., 2022).

3 STATISTICAL LIMITS OF OPTIMAL SUBSET SELECTION

In this section, we start by defining the task of finding the optimal subset in the training set. A natural comparator (or goal) is as follows: Fix a possibly randomized learning algorithm $\mathcal{A} = \{\mathcal{A}_n\}_{n\in\mathbb{N}}$ and a training sample $S = (Z_1, \ldots, Z_n) \sim P^n$. Assume that we have a budget of size m for data selection. For every index set $I \subseteq [n]$, let $S_I = (Z_i)_{i\in I}$, with the indices in increasing order. Let

$$I_m^\star \in \operatorname*{arg\,min}_{\substack{I\subseteq[n]\\ |I|\le m}} R_P(\mathcal{A}_m(S_I))\,, \tag{1}$$

with ties broken canonically. Thus $R_P(\mathcal{A}_m(S_{I_m^\star}))$ is the risk of the best subset that an oracle with access to P can find in the realized sample. Importantly, the oracle must use the same learner $\mathcal{A}$ as the data selector. Recall from Section 2.1 that $R_P^\star$ denotes the Bayes-optimal population risk.

Let $V \sim P^v$ be an independent validation set, and let a possibly randomized data selection algorithm be a map $\Pi : \mathcal{Z}^n \times \mathcal{Z}^v \longrightarrow \{I \subseteq [n] : |I| \le m\}$. Write $\widehat{I}_m := \Pi(S, V)$. The selected excess risk decomposes as

$$R_P\Big(\mathcal{A}_m(S_{\widehat{I}_m})\Big) - R_P^\star = \underbrace{R_P\big(\mathcal{A}_m(S_{I_m^\star})\big) - R_P^\star}_{\text{approximation error}} + \underbrace{R_P\Big(\mathcal{A}_m(S_{\widehat{I}_m})\Big) - R_P\big(\mathcal{A}_m(S_{I_m^\star})\big)}_{\text{selection error}}. \tag{2}$$

Both terms are nonnegative: the first asks whether the training sample contains a good size-m subset for $\mathcal{A}_m$, and the second measures the cost of finding one using Π. As we will show there exists class of problems such that we have a dichotomy: either even the best subset for $\mathcal{A}_m$ makes the approximation error large, or a good subset exists but Π cannot identify it.

Theorem 3.1. *Consider multiclass classification with ordinary 0–1 loss. Fix any compression fraction $\lambda \in (0,1)$, and set $K := 2^{\lceil 100/(1-\lambda)\rceil}$. Let the label space be $[K]$. There exist $n_\lambda \in \mathbb{N}$ and $C_\lambda > 0$, depending only on λ, with the following property.*

For every $n \ge n_\lambda$, set $m = \lfloor \lambda n \rfloor$. For every integer validation size $v \ge 0$, there is a finite family $\mathfrak{P}_{\lambda,n,v}$ of deterministic-label K-class classification distributions, together with an explicit deterministic learner $\mathcal{A}_{\mathrm{lin}}$. Every member of the family has Bayes risk zero.

For every pair consisting of a learner $\mathcal{A}_m$ and a data selector Π, there exists $P \in \mathfrak{P}_{\lambda,n,v}$ such that, for independent samples $S \sim P^n$ and $V \sim P^v$,

$$\begin{aligned}&\mathbb{E}_S\big[R_P\big(\mathcal{A}_m(S_{I_m^\star})\big) - R_P^\star\big]\\ &+ \mathbb{E}_{S,V,\Pi}\Big[R_P\Big(\mathcal{A}_m(S_{\widehat{I}_m})\Big) - R_P\big(\mathcal{A}_m(S_{I_m^\star})\big)\Big]\\ &\ge \frac{\lambda(1-\lambda)^2}{10^6}\frac{n}{n+v}.\end{aligned} \tag{3}$$

Here $\widehat{I}_m = \Pi(S,V)$ satisfies $|\widehat{I}_m| \le m$, and $I_m^\star$ minimizes $R_P(\mathcal{A}_m(S_I))$ over $I \subseteq [n]$ with $|I| \le m$.

Moreover, these families and the corresponding learners can be chosen so that, for every $n \ge n_\lambda$ and for every $P \in \mathfrak{P}_{\lambda,n,v}$

$$\frac{n+v}{n}\,\mathbb{E}_S[R_P(\mathcal{A}_{\mathrm{lin}}(S)) - R_P^\star] \le C_\lambda n^{-1/5}, \tag{4}$$

$$\frac{n+v}{n}\,\mathbb{E}_S\big[R_P\big(\mathcal{A}_{\mathrm{lin}}(S_{I_m^\star})\big) - R_P^\star\big] \le C_\lambda n^{-1/5}. \tag{5}$$

In Equation (5), $I_m^\star$ is the optimal subset for $\mathcal{A}_{\mathrm{lin}}$.

Let us unpack the implications of this result:

Accurate subsets exist: Bayes risk is zero, and $\mathcal{A}_{\mathrm{lin}}$ achieves vanishing error on both the full sample and a distribution-aware oracle subset of at most $\lfloor \lambda n \rfloor$ examples.

Selection is hard: Finding such a subset can nevertheless be statistically hard: for fixed λ and $v = O(n)$, every selector-algorithm pair incurs a nonvanishing worst-case selection error, even with unlimited computation.

Universality: every learner–selector pair has worst-case approximation plus selection error at least $\Omega_\lambda(n/(n+v))$. Consequently, uniformly vanishing total excess risk requires $v/n \to \infty$, meaning that the number of validation sample should grow faster than n

Tightness: Our lower bound in Theorem 3.1 is tight in its dependence on the validation size v. Specifically, conditional on the training sample S, running $\mathcal{A}_m$ on all

subsets of S of size at most m induces a finite hypothesis class of size at most $\binom{n}{m} \le (en/m)^m$. Standard generalization bounds for empirical risk minimization over a finite class (Shalev-Shwartz and Ben-David, 2014) guarantee excess selection error $O\Big(\frac{m\log(n/m)}{v}\Big)$, meaning a validation set of size $v = \Theta\Big(\frac{m\log(n/m)}{\varepsilon}\Big)$ suffices to achieve selection error ε. This matches our lower bound up to a constant factor depending only on λ.

Remark 3.2 (Relation to long-tail memorization)**.** *The data distribution used in our proof shares the sparse $O(1/n)$ subpopulation occupancy of the long-tail constructions introduced in the context of data memorization by Feldman (2020), where many subpopulations appear zero or one time in the training set. The crucial technical difference in our construction is the graph structure connecting different subpopulations. In Feldman (2020), subpopulations are independent, so any subpopulation with zero training samples incurs constant error even for full-data training and for a population-aware oracle. In our construction, the local vertex-parity relations on the graph allow both full-data training and the budget-m oracle to select a few neighboring observations from S and accurately predict the label of an active test subpopulation even when that subpopulation has zero samples in the training set.*

Remark 3.3 (Connection to practice)**.** *Our construction of hard data distribution, while stylized, captures two central empirical properties of modern large-scale datasets such as those used in LLM pretraining. First, pretraining data follows a heavy long-tailed distribution containing vast numbers of rare facts or specialized code patterns that appear only once or a few times in the training set (Feldman, 2020). Second, unlike prior theoretical models that treat rare subpopulations as completely independent (Feldman, 2020), real-world tail tasks exhibit* task relatedness*: even if a specific test query or fact is missing from the training corpus, a model trained on the full dataset can often deduce the answer by combining related documents or code dependencies. This task relatedness is precisely what allows a small optimal subset $S_{I^\star_m}$ to exist: an oracle that knows which tail capabilities will be tested can retain just the few supporting "neighbor" examples needed to reconstruct them. However, because rare capabilities are seldom probed in a moderate-sized validation split V, no sample-based selector can tell which parts of the tail will be queried at test time, making it impossible to prune the corpus to budget m.*

Proof overview of Theorem 3.1. We first describe the main challenge in proving Theorem 3.1 and the high-level idea behind our construction. The natural starting point is to consider a long-tailed distribution with many rare subpopulations so that accurate prediction requires retaining a large amount of information from the training sample. The immediate difficulty is that if predicting well requires storing more than m pieces of information, then even full-data training and a population-aware oracle with budget m will fail. Our construction resolves this difficulty through two main ideas. First, we include a shared baseline task that every learner must solve at test time; both the oracle and any data selector must spend almost all of their subset budget m on this baseline task, leaving only a small residual budget for the remaining subpopulations. Second, we connect the remaining subpopulations through a sparse graph where each subpopulation's label is determined by the labels of its immediate neighbors:

1. *Hidden test relevance.* Most of the time, all graph subpopulations are sampled at the same rate in an *easy* form where the input features directly reveal the label. This allows the training set to teach the learner the labels of the observed subpopulations, while ensuring that any subpopulation dropped by the oracle incurs zero error when it appears in this easy form at test time. However, for a small fraction of *active* subpopulations, the distribution also asks *hard* test queries where the label is not visible in the input features and must be predicted from what the learner retained during training. Because these hard queries are rare, most graph groups receive none in the training or validation sets. An oracle that knows the test distribution knows which subpopulations are active and can spend its small residual budget solely on them, whereas a data selector cannot tell which subpopulations will be tested as hard queries.

2. *Recovering unseen subpopulations via graph connections.* Because there are as many subpopulations as training examples, a constant fraction of the active subpopulations receive zero samples in the training set. If the subpopulations were independent, neither the full dataset nor the oracle could predict these unseen subpopulations at test time. Because the subpopulations are linked on a sparse graph without short loops, however, a typical unseen active subpopulation lies, with high probability, in a small missing component whose boundary subpopulations *did* appear in the training set. By retaining only those few boundary examples, the oracle (as well as full-data training) can deduce the exact label of the active subpopulation. A data selector, not knowing which subpopulations are active, would have to preserve such boundary examples across the entire graph, which requires far more examples than its budget allows.

□

4 FAITHFUL DATA SELECTION

The previous section studied the conventional goal of data selection: returning a subset of the training sample whose population error nearly matches that of the best subset $I_m^\star$ in Equation (1). However, our results show that for every pair consisting of a learning algorithm and a data selection algorithm, there exists a scenario in which the data selection algorithm fails miserably unless it has $\Omega(n)$ validation samples, even though a simple learner trained on all of S succeeds there. We therefore change the goal.

The selection error in Equation (2) compares the chosen subset with $I_m^\star$, which depends on P and is therefore unknown to the selector. We instead split the same excess risk against a comparator that the selector can access, the model trained on all of S. Precisely, write

$$R_P\big(\mathcal{A}(S_{\widehat{I}})\big) - R_P^\star = \underbrace{R_P\big(\mathcal{A}(S)\big) - R_P^\star}_{\text{full-data excess risk}} + \underbrace{R_P\big(\mathcal{A}(S_{\widehat{I}})\big) - R_P\big(\mathcal{A}(S)\big)}_{\text{faithfulness gap}}. \quad (6)$$

The first term does not involve the selector and is typically small in practice, since data selection is usually applied when training on the full sample already performs well. The excess risk of the selected subset is then essentially its faithfulness gap, so we seek a subset on which training yields a model that closely tracks the full-data model.

Remark 4.1. *We do not claim that we can overcome the fundamental lower bound in Theorem 3.1, since the result provides a lower bound the excess error directly. Thus, for that class of distribution the faithfulness gap is still large. However, as we show next, we provide a practical algorithm with certification: if the certificate is small, then, we know that the faithfulness gap is small. Also, the proposed certificate can be computed only based on the information of the training set.*

4.1 From faithfulness to KL divergence

How can we control the faithfulness gap? The faithfulness gap in Equation (6) is the difference between the averages of one function, $\theta \mapsto R_P(\theta)$, under two distributions over trained parameters: the law $\mathcal{A}_n(S)$ of the model trained on S and the law $\mathcal{A}_m(S_{\widehat{I}})$ of the model trained on the subset. We cannot evaluate this function, since P is unknown, but we do not need to: if the two *laws* are close, every bounded function of θ has nearly the same average under both, whatever P is. We therefore call a subset *faithful* if training on it produces nearly the same distribution over models as training on S. This requirement is stronger than a small faithfulness gap, and usefully so: it controls the gap under every data distribution at once, and it controls every other bounded evaluation of the trained model, which we call a *query*. Examples include accuracy on a subgroup or on a rare subpopulation, accuracy under distribution shift, and the loss or prediction at a single test point as in data attribution (Ye et al., 2024). The next result, whose proof is simply based on the variational representation of total variation distance and Pinsker's inequality, formalizes this intuition:

Proposition 4.2. *Let $a, b \in \mathbb{R}$ and $n, m \in \mathbb{N}$ such that $1 \le m \le n$. Let $q : \mathbb{R}^d \to [a, b]$ be a bounded-function. Fix $S \in \mathcal{Z}^n$ as a training set of size n and let $I \subseteq [n]$ be an arbitrary subset of $[n]$ such that $|I| = m$. Then, for every arbitrary randomized algorithm $\mathcal{A} = \{\mathcal{A}_m\}_{m\in\mathbb{N}}$, we have error satisfies*

$$\Big|\, \mathbb{E}_{\theta\sim\mathcal{A}_n(S)}[q(\theta)] - \mathbb{E}_{\theta'\sim\mathcal{A}_m(S_I)}[q(\theta')] \,\Big| \le \sqrt{\frac{(b-a)^2}{2}\, D_{\mathrm{KL}}\big(\mathcal{A}_T(S) \,\|\, \mathcal{A}_T(I)\big)}.$$

5 PRACTICAL DATA SELECTION USING LANGEVIN ALGORITHM

Proposition 4.2 reduces faithfulness to a single number, $D_{\mathrm{KL}}\big(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I)\big)$. How can we turn this result to a faithful data selection algorithm with certificate guanratee? To select subsets with it, we need two things: an upper bound on this divergence that can be evaluated based on the training set, and a form of that bound that can be minimized over subsets. For a general learner, neither is available, since the laws $\mathcal{A}_n(S)$ and $\mathcal{A}_m(S_I)$ have no explicit form. We therefore approximate gradient-based training by the Unadjusted Langevin Algorithm (ULA), under which both become possible: we bound the divergence in closed form and then turn the bound into a practical and scalable algorithm.

Fix a number of iterations $T \ge 1$, step sizes $h_0, \dots, h_{T-1} > 0$, inverse temperatures $\beta_0, \dots, \beta_{T-1} > 0$, and an initial law μ_0. On a training set S', ULA draws $\theta_0 \sim \mu_0$ and, for $t = 0, \dots, T-1$, let

$$\theta_{t+1} = \theta_t - h_t \nabla L_{S'}(\theta_t) + \sqrt{2h_t/\beta_t}\, \xi_{t+1}, \quad (7)$$

where $\xi_1, \xi_2, \dots$ are i.i.d. $\mathcal{N}(0, I_d)$ and independent of θ_0. We write $\mu_t^{S'}$ for the law of θ_t, and we take the learner to be ULA stopped at iteration T, that is, $\mathcal{A}_k(S') := \mu_T^{S'}$ for every sample S' of size k. ULA is gradient descent with injected Gaussian noise, and its stochastic-gradient version is the SGLD algorithm of Welling and Teh (2011). The noise is what makes faithfulness measurable. ULA is an approximation of how models are trained in practice. Our guarantees, including the certificate, are proven for ULA, and Section 6 shows

that the resulting selection rule remains effective when models are trained with standard optimizers such as SGD and Adam.

Before stating our results, we define the weights $w_t := \beta_t h_t/4$ and a quantity that we refer to as the *trajectory gradient discrepancy*,

$$G_T(I) := \sum_{t=0}^{T-1} w_t \, \mathbb{E}_{\theta\sim\mu_t^S} \big\|\nabla L_{S_I}(\theta) - \nabla L_S(\theta)\big\|^2, \quad (8)$$

the squared distance between the subset and full-data gradients, accumulated along the full-data training trajectory with weight w_t at iteration t.

Theorem 5.1. *Fix $T \geq 1$, step sizes $h_0, \ldots, h_{T-1} > 0$, inverse temperatures $\beta_0, \ldots, \beta_{T-1} > 0$, an initial law μ_0, integers $1 \leq m \leq n$, a training set $S \in \mathcal{Z}^n$, and $I \subseteq [n]$ with $|I| = m$, and assume that $\ell(\cdot, z)$ is continuously differentiable for every $z \in \mathcal{Z}$. Run ULA on S and on S_I with these schedules and the same initial law μ_0. Then, $D_{\mathrm{KL}}\big(\mathcal{A}_n(S) \,\|\, \mathcal{A}_m(S_I)\big) \leq G_T(I)$*

Then, combined with Proposition 4.2, Theorem 5.1 turns $G_T(I)$ into a certificate of faithfulness.

Corollary 5.2 (Faithfulness certificate under ULA)**.** *Under the assumptions of Theorem 5.1, for all $a < b$ and every measurable $q : \mathbb{R}^d \to [a, b]$,*

$$\Big|\, \mathbb{E}_{\theta\sim\mathcal{A}_n(S)}[q(\theta)] - \mathbb{E}_{\theta\sim\mathcal{A}_m(S_I)}[q(\theta)] \,\Big| \leq (b-a)\sqrt{G_T(I)/2}.$$

In particular, under every data distribution P, the faithfulness gap measured in any loss with values in $[0, 1]$ is at most $\sqrt{G_T(I)/2}$.

This result shows that $G_T(I)$ can be thought of as a principled selection objective. A notable property is that the expectation in equation 8 is over the full-data chain only, so $G_T(I)$ can be estimated for every candidate subset from the same full-data training runs, using per-example gradients; this requires no training on any subset and no access to P or to held-out data.

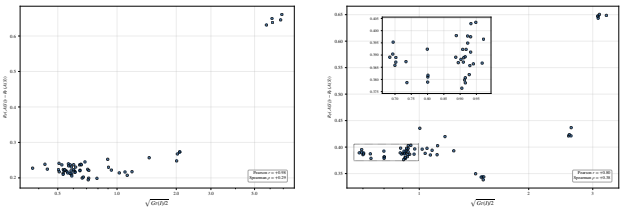
Remark 5.3. *MH:Mehdi please fill it traceIN gradmatch*

Theorem 5.4 (Selection as a quadratic program)**.** *In the setting of Theorem 5.1, write $\zeta_i(\theta) := \nabla\ell(\theta, Z_i) - \nabla L_S(\theta)$ for the centered per-example gradients, assume that $\mathbb{E}_{\theta\sim\mu_t^S}\|\nabla\ell(\theta, Z_i)\|^2 < \infty$ for all t and i, and define $C \in \mathbb{R}^{n\times n}$ by*

$$C_{ij} := \sum_{t=0}^{T-1} w_t \, \mathbb{E}_{\theta\sim\mu_t^S}\big\langle \zeta_i(\theta), \zeta_j(\theta)\big\rangle.$$

Then C is positive semidefinite, $C\mathbf{1} = 0$, and $G_T(I) = \frac{1}{m^2}\mathbf{1}_I^\top C\,\mathbf{1}_I$ for every $I \subseteq [n]$ with $|I| = m$. Consequently, minimizing $G_T(I)$ over subsets of size m is the integer quadratic program

$$\min_{\alpha\in\mathcal{B}_m} G(\alpha), \qquad G(\alpha) := \frac{1}{m^2}\alpha^\top C\alpha, \qquad \mathcal{B}_m := \big\{\alpha \in \{0,1\}^n : \mathbf{1}^\top\alpha = m\big\}$$



(a) Resnet18 on CIFAR10 (b) VGG-16 on CIFAR100

Figure 1: Faithfulness gap $R_P(\mathcal{A}(S_I)) - R_P(\mathcal{A}(S))$ versus the certificate $\sqrt{G_T(I)/2}$ for CIFAR-10 / ResNet-18 (left) and CIFAR-100 / VGG-16 (right) at retention fraction 0.1. Each point is represents a subset from a data selection method, and $\mathcal{A}(S)$ represents SGD. Subsets with large $G_T(I)$ are the ones with the largest gap. The inset (right) zooms on the dotted region. Other settings are plotted in Section H

5.1 Subset selection via gradient swaps

How can we solve the program in Theorem 5.4 when n is large? Three facts rule out the obvious approaches. First, exact minimization is NP-hard. Second, forming C requires $\Theta(n^2)$ inner products at every point of the trajectory, and storing it requires $\Theta(n^2)$ memory. Third, relaxing α to $[0, 1]^n$ is uninformative: since $C\mathbf{1} = 0$, the fractional point $\frac{m}{n}\mathbf{1}$ has objective value zero and says nothing about which subset to select. We therefore use a local search over $\mathcal{B}_m$ that accesses C only through matrix–vector products and certifies how far its objective value is from the minimum.

Theorem 5.5 (Swap descent decreases the faithfulness bound)**.** *In the setting of Theorem 5.4, let $1 \leq m \leq n-1$, and for $\alpha = \mathbf{1}_I \in \mathcal{B}_m$ write $D(\alpha) := G_T(I)$. For every initial point $\alpha_0 \in \mathcal{B}_m$ and every tie-breaking rule in the choice of the swapped pair, ?? generates iterates $\alpha_0, \alpha_1, \ldots \in \mathcal{B}_m$ and returns $\alpha_{\mathrm{out}} = \mathbf{1}_{I_{\mathrm{out}}}$ such that:*

(i) every accepted swap strictly decreases the objective, $D(\alpha_{s+1}) < D(\alpha_s)$, and the algorithm terminates after finitely many iterations;

(ii) if α_0 is drawn uniformly from $\mathcal{B}_m$, then

$$\mathbb{E}\big[D(\alpha_{\mathrm{out}})\big] \leq \frac{n-m}{m(n-1)} \sum_{t=0}^{T-1} w_t V_t,$$

where $V_t := \frac{1}{n}\sum_{i=1}^n \mathbb{E}_{\theta\sim\mu_t^S}\|\nabla\ell(\theta, Z_i) - \nabla L_S(\theta)\|^2$ and the expectation is over the initialization.

6 EXPERIMENTS

We compare seven data-selection baselines on CIFAR10 and CIFAR100 (Alex, 2009) using three architectures: VGG-16 (Simonyan and Zisserman, 2015), Resnet-18

(He et al., 2016), and Inception (Szegedy et al., 2014). Every baseline selects four retention fractions of the training pool (10%, 20%, 40%, 50%), then the model is retrained on the selected set. Test accuracy, averaged over 5 runs is reported as an evaluation metric.

We compare our approach to the following data selection baselines:

- **Herding** (Welling, 2009), **Moderate Coreset** (Xia et al., 2022), **Facility Location** and **GraphCut** (Iyer et al., 2021) are one-shot, embedding-based methods; each reads penultimate-layer features ϕ from a *single* converged reference checkpoint. Facility Location and GraphCut use the `apricot` lazy-greedy submodular optimizer with a Euclidean similarity on ϕ.
- **EL2N** (Paul et al., 2021) TO EXPLAIN HERE
- **Random** draws a uniform subset of the training pool independently per seed, with no selection cost.

6.1 Checkpointing

Running **??** still requires $O(nTd)$ computation, therefore making it expensive for any reasonably large architecture, dataset and number of iterations.

We would like to see whether not storing all optimization checkpoints (selecting using only information for the first 5 epochs for example) yields good selection.

6.2 Subset similarity across architectures

Also whether using smaller architectures maintains the selection quality while reducing the computation cost.

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In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, "we checked LLM-generated research ideas for potential plagiarism through a manual literature survey", "LLM-generated code was verified and tested for correctness by 2 authors", etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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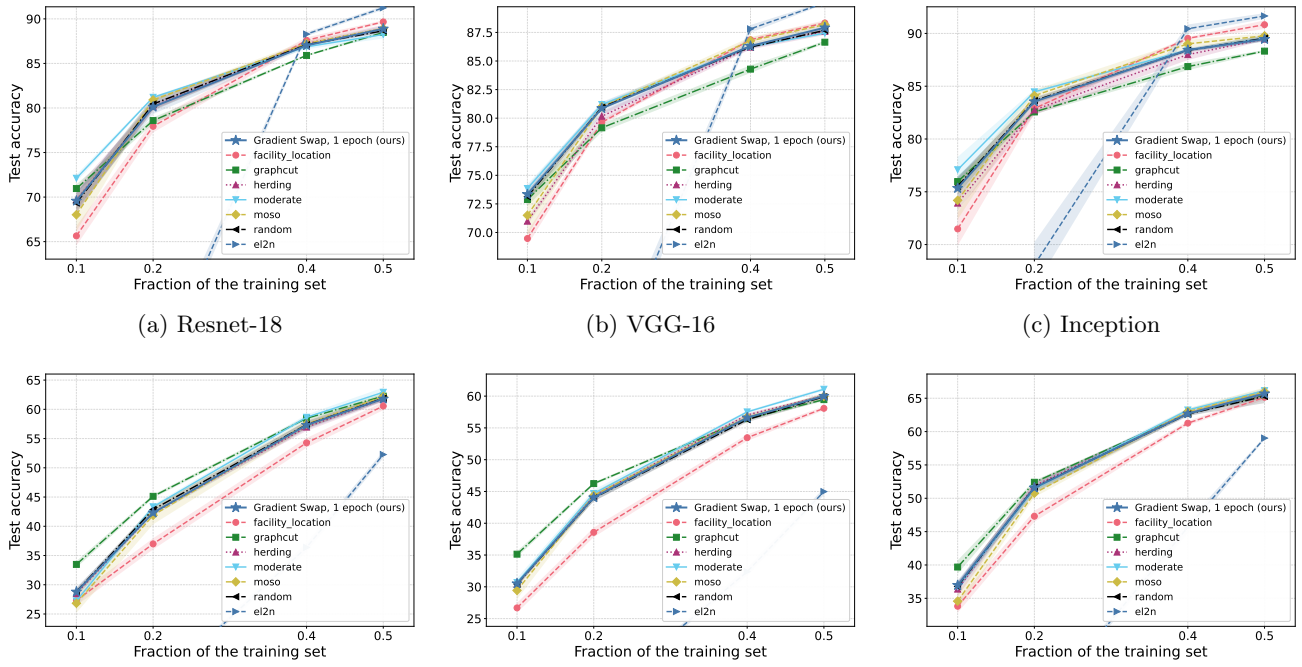
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(a) Resnet-18 (b) VGG-16 (c) Inception

(d) Resnet-18 (e) VGG-16 (f) Inception

Figure 2: Comparing our method against data selection baselines, on CIFAR10 (Figures 2a to 2c) and CIFAR100 (Figures 2d to 2f), averaged over 5 seeds. Our method matches state of the art static data selection baselines while providing a certificate of optimality

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submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement.

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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Towards Certifiable Data Selection: Statistical Limits and Scalable Algorithms: Supplementary Materials

A APPENDIX OF

Proof of Theorem 3.1. We first define the distribution family and a prior on its hidden parameters, then prove the universal lower bound. We define the explicit learner only afterward and show that both full-data training and the population-aware oracle achieve vanishing error.

Data distribution. Fix $\lambda \in (0,1)$, let $m = \lfloor \lambda n \rfloor$, and write $\gamma = 1 - \lambda$. Define

$$\begin{aligned} B &:= \left\lceil \frac{100}{\gamma} \right\rceil, \qquad K := 2^B, \qquad \kappa := \lambda - \frac{\lambda\gamma}{10}, \qquad p_{\mathrm{c}} := \lambda - \frac{\lambda\gamma}{20}, \\ g &:= \left\lceil \frac{100}{\lambda} \right\rceil, \qquad D := \lfloor \kappa n \rfloor, \qquad \omega := \frac{\gamma}{100g} \frac{n}{n+v}, \qquad a := \frac{1 - p_{\mathrm{c}} - \omega\kappa}{1+\omega}. \end{aligned} \tag{9}$$

The four masses $a, p_{\mathrm{c}}, \omega a, \omega\kappa$ are positive and sum to one. We will use

$$\frac{99\gamma}{100} \le a \le \frac{21\gamma}{20}, \qquad \frac{1}{2} < a + \kappa < 1, \qquad p_{\mathrm{c}} - \kappa = \frac{\lambda\gamma}{20}, \qquad \lambda - \kappa = \frac{\lambda\gamma}{10}. \tag{10}$$

Indeed, if $b = 1 - p_{\mathrm{c}} = \gamma + \lambda\gamma/20$, then

$$b - a = \frac{\omega(b+\kappa)}{1+\omega} \le \omega \le \frac{\gamma}{100}, \qquad b + \kappa < 1,$$

which gives the displayed bounds.

The public graph. We use the following quantitative consequence of the Erdős–Sachs construction: for every integer $L \ge 3$, there is a simple cubic graph H_L with girth at least L and

$$|V(H_L)| \le 4 \sum_{i=1}^{L-2} 2^i = 2^{L+1} - 8, \qquad |E(H_L)| \le 3 \cdot 2^L - 12 \tag{11}$$

For sufficiently large n, choose

$$L := \left\lfloor \log_2\!\left(\frac{\sqrt{n}}{3g} \right) \right\rfloor \ge 3. \tag{12}$$

Take g disjoint copies of H_L to form a block with $b_n = g|E(H_L)| \le \sqrt{n}$ edges. Let $G = G_{\lambda,n,v}$ be the disjoint union of $\lfloor an/b_n \rfloor$ such blocks, and let $N = b_n \lfloor an/b_n \rfloor$ be its number of edges. Then

$$g \mid N, \qquad N \le an, \qquad 0 \le an - N < \sqrt{n}, \qquad \mathrm{girth}(G) \ge L. \tag{13}$$

Moreover,

$$L \ge \frac{1}{2} \log_2 n - \log_2(3g) - 1. \tag{14}$$

The block and L depend only on (λ, n); only the number of copies depends on v through a. All bounds are uniform in v, and $N \ge (99\gamma/100)n - \sqrt{n}$ for sufficiently large n.

Partition $E(G)$ into $J = N/g$ disjoint ordered groups $\mathcal{G}_1, \ldots, \mathcal{G}_J$, each containing g edges, and write $e(j,\ell)$ for the ℓ-th edge of $\mathcal{G}_j$.

Hidden parameters. Each member of the family is indexed by a triple (Θ, Ψ, h):

1. The graph class vectors $\Theta = (\Theta_e)_{e \in E(G)}$, with $\Theta_e \in \mathbb{F}_2^B$, satisfy the coordinatewise vertex parities
$$\Theta_{e_1} + \Theta_{e_2} + \Theta_{e_3} = 0 \quad \text{whenever } e_1, e_2, e_3 \text{ meet at a vertex.} \tag{15}$$
2. The active indices $h = (h_1, \ldots, h_J) \in [g]^J$ designate one active edge $e(j, h_j) \in \mathcal{G}_j$ in each group.
3. The compositional matrix $\Psi \in \mathbb{F}_2^{D \times B}$ associates each descriptor $q \in \mathbb{F}_2^D$ with the vector $q^\top \Psi \in \mathbb{F}_2^B$. Writing b_i for the i-th standard basis vector, $b_i^\top \Psi$ is the i-th row of Ψ.

Graph identifiers and compositional identifiers are tagged to distinguish them. For either type of domain u, set
$$\theta_u := \begin{cases} \Theta_e, & u = e \in E(G), \\ q^\top \Psi, & u = q \in \mathbb{F}_2^D. \end{cases}$$
Fix a public bijection $\mathrm{cl} : \mathbb{F}_2^B \to [K]$, and define the true class $c_u := \mathrm{cl}(\theta_u)$.

Observation regimes. Every example has feature
$$X = (u, z) \in \left(E(G) \dot\cup \mathbb{F}_2^D\right) \times \left([K] \cup \{\perp\}\right)$$
and deterministic label $Y = c_u$. A draw is generated according to two observation regimes:

- **Case 1: Clear observations (total probability $a + p_c$).** The feature directly reveals the class:
$$X = (u, c_u), \qquad Y = c_u.$$
There are two sub-cases:
  1. With probability a, choose a graph edge $u = e$ uniformly from $E(G)$, so each edge has probability a/N.
  2. With probability p_c, choose a compositional descriptor $u = q$ uniformly from $\mathbb{F}_2^D$, so each descriptor has probability $p_c/2^D$.
- **Case 2: Missing-cue observations (total probability $\omega(a + \kappa)$).** The cue is erased:
$$X = (u, \perp), \qquad Y = c_u.$$
Again there are two sub-cases:
  1. With probability ωa, choose j uniformly from $[J]$ and set $u = e(j, h_j)$. Thus each active edge has probability $\omega a/J = \omega a g/N$, while inactive edges have probability zero.
  2. With probability $\omega\kappa$, choose i uniformly from $[D]$ and set $u = b_i$, so each basis vector has probability $\omega\kappa/D$.

Denote the resulting distribution by $P_{\Theta,\Psi,h}$. Training, validation, and test examples are conditionally i.i.d. from this same distribution. The rule
$$f^\star_{\Theta,\Psi}(u, z) := c_u \tag{16}$$
labels every point in the support, and hence
$$R^\star_{P_{\Theta,\Psi,h}} = 0. \tag{17}$$
The finite family is
$$\mathfrak{P}_{\lambda,n,v} := \{P_{\Theta,\Psi,h} : (\Theta, \Psi, h) \text{ is admissible}\}.$$

Meta-distribution. Fix an arbitrary learner $\mathcal{A}$ and selector Π. To prove the universal lower bound, place a product prior on the finite parameter space indexing $\mathfrak{P}_{\lambda,n,v}$. All entropies below are in bits. Draw the three collections independently as follows:

1. *Graph classes.* Each of the B columns of Θ is uniform in the binary cycle space

$$\mathcal{C}(G) := \left\{ x \in \mathbb{F}_2^{E(G)} : x_{e_1} + x_{e_2} + x_{e_3} = 0 \text{ at every vertex} \right\}.$$

   Since a cubic graph with N edges has $2N/3$ vertices,

$$\dim_{\mathbb{F}_2} \mathcal{C}(G) = N - \frac{2N}{3} + \#\{\text{connected components of } G\} \geq \frac{N}{3}.$$

   Thus $H(\Theta) \geq NB/3$ bits.

2. *Compositional classes.* The DB entries of Ψ are independent Bernoulli(1/2) bits, so $H(\Psi) = DB$.

3. *Active edge indices.* The indices $h_1, \ldots, h_J$ are independent and uniform in $[g]$.

Reduction to conditional entropy. Let W contain the observation regimes and domain identifiers of every example in $S \cup V$, together with the independent random seeds of Π and $\mathcal{A}$, but neither the class labels nor the clear cues. Then W is independent of (Θ, Ψ), although missing-cue graph identifiers can reveal some coordinates of h.

Let $U \subseteq [J]$ be the groups receiving no missing-cue graph observation in $S \cup V$, and set

$$E_U := \bigcup_{j \in U} \mathcal{G}_j.$$

We regard Θ_{E_U} as an N-coordinate vector padded with zeros outside E_U. Both U and E_U are determined by W. Each group is unresolved with probability

$$\begin{aligned} s_n := \mathbb{P}\{j \in U\} &= \left(1 - \frac{\omega a g}{N}\right)^{n+v} \\ &\geq 1 - \frac{(n+v)\omega a g}{N} = 1 - \frac{\gamma a n}{100N} \geq \frac{9}{10} \end{aligned} \tag{18}$$

for all sufficiently large n, uniformly in v. If $j \in U$, then h_j remains uniform in $[g]$, independently of (Θ, Ψ), the entire labeled sample (S, V), and the final predictor, conditional on W. Indeed, the probability of receiving no such query is the same for every value of h_j, and clear observations do not depend on h_j.

Let $\widehat{I}_m = \Pi(S, V)$ and write

$$F := \mathcal{A}(S_{\widehat{I}_m})$$

for the resulting predictor. For the missing-cue inputs define

$$\varepsilon_e := \mathbf{1}\{F(e, \perp) \neq c_e\}, \quad e \in E_U, \qquad \varepsilon_i := \mathbf{1}\{F(b_i, \perp) \neq c_{b_i}\}, \quad i \in [D],$$

and let

$$Q := \sum_{e \in E_U} \varepsilon_e + \sum_{i=1}^{D} \varepsilon_i.$$

For every unresolved group, averaging over the still-uniform active index assigns each of its g edges missing-cue test mass $\omega a/(Jg) = \omega a/N$. Each basis vector has missing-cue mass $\omega\kappa/D$. Discarding all other nonnegative loss terms and using $N \leq an$ and $D \leq \kappa n$ gives

$$\begin{aligned} \mathbb{E} R_{P_{\Theta,\Psi,h}}(F) &\geq \frac{\omega a}{N} \mathbb{E} \sum_{e \in E_U} \varepsilon_e + \frac{\omega\kappa}{D} \mathbb{E} \sum_{i=1}^{D} \varepsilon_i \\ &\geq \frac{\omega}{n} \mathbb{E} Q. \end{aligned} \tag{19}$$

All expectations in this proof include the parameter prior, data, and algorithmic randomness.

Encode the selected sample by a message

$$M := \left(\mathbf{1}_{\widehat{I}_m}, (c_{u_i})_{i\in\widehat{I}_m}\right),$$

padding the class list to m entries. Given (W, M), every selected input and label can be reconstructed: for a clear observation its class determines both its cue and label, while for a missing-cue observation W supplies the input and M supplies the label. Because the learner seed belongs to W, the predictor F is determined by (W, M).

Given (W, M), evaluate F at every missing-cue domain in $E_U \cup \{b_1, \ldots, b_D\}$. The target (Θ_{E_U}, Ψ) can then be reconstructed by specifying an error mask of length at most $N + D$ and the true B-bit vector at each of the Q mistaken domains. Correct predictions require no additional bits. Therefore,

$$H(\Theta_{E_U}, \Psi \mid W, M) \le (N + D) + B\,\mathbb{E}Q \le n + B\,\mathbb{E}Q, \tag{20}$$

where $N + D \le (a + \kappa)n < n$. Combining Equations (19) and (20) yields

$$\mathbb{E}R_{P_{\Theta,\Psi,h}}(F) \ge \frac{\omega}{nB}\left(H(\Theta_{E_U}, \Psi \mid W, M) - n\right). \tag{21}$$

Computing the conditional entropy. The chain rule gives

$$\begin{aligned} H(\Theta_{E_U}, \Psi \mid W, M) &= H(\Theta_{E_U}, \Psi \mid W) - I((\Theta_{E_U}, \Psi); M \mid W) \\ &\ge H(\Theta_{E_U}, \Psi \mid W) - H(M \mid W). \end{aligned} \tag{22}$$

First, the n-bit selected-index indicator and at most m class labels, each represented by B bits, determine M. This accounts for information encoded through the choice of indices, including information extracted from the validation labels, and gives

$$H(M \mid W) \le n + mB. \tag{23}$$

Second, $\mathbb{E}|E(G) \setminus E_U| \le N/10$ by Equation (18). Since W is independent of (Θ, Ψ), restricting Θ to E_U removes at most B bits per discarded edge. Hence

$$H(\Theta_{E_U} \mid W) \ge H(\Theta \mid W) - B\,\mathbb{E}|E(G) \setminus E_U| \ge \left(\frac{1}{3} - \frac{1}{10}\right) NB = \frac{7N}{30}B.$$

The matrix Ψ contributes DB independent bits, so

$$H(\Theta_{E_U}, \Psi \mid W) \ge \left(D + \frac{7N}{30}\right) B. \tag{24}$$

Substituting Equations (23) and (24) into Equation (22) gives

$$H(\Theta_{E_U}, \Psi \mid W, M) \ge \left(D + \frac{7N}{30} - m\right) B - n. \tag{25}$$

After division by nB, the right-hand side is at least

$$\begin{aligned} \kappa + \frac{7a}{30} - \lambda - \frac{1}{B} - \frac{1}{n} - \frac{7}{30\sqrt{n}} &\ge -\frac{\gamma}{10} + \frac{7}{30}\frac{99\gamma}{100} - \frac{\gamma}{100} - \frac{1}{n} - \frac{7}{30\sqrt{n}} \\ &= \frac{121\gamma}{1000} - \frac{1}{n} - \frac{7}{30\sqrt{n}} \ge \frac{\gamma}{10} \end{aligned}$$

for all sufficiently large n, depending only on λ. Thus

$$H(\Theta_{E_U}, \Psi \mid W, M) \ge \frac{\gamma nB}{10}. \tag{26}$$

Completing the lower bound. Applying Equation (26) in Equation (21), and using $1/B \le \gamma/100$, gives

$$\begin{aligned}\mathbb{E}R_{P_{\Theta,\Psi,h}}(F) &\ge \frac{\omega}{nB}\left(\frac{\gamma nB}{10} - n\right) \ge \frac{9\omega\gamma}{100} \\ &= \frac{9\gamma^2}{10000g}\frac{n}{n+v} \ge \frac{\lambda(1-\lambda)^2}{10^6}\frac{n}{n+v},\end{aligned} \tag{27}$$

where the last inequality uses $g = \lceil 100/\lambda \rceil \le 101/\lambda$.

This is an average over a finite prior, so some fixed $P_{\Theta,\Psi,h} \in \mathfrak{P}_{\lambda,n,v}$ attains the same lower bound. Since its Bayes risk is zero, the approximation and selection terms in Equation (3) telescope to this selected population risk. This proves the universal lower bound. It remains to construct the learner promised in the theorem and prove its two recovery guarantees.

The explicit learner. The learner $\mathcal{A}_{\mathrm{lin}}$ depends only on the public family, not on the arbitrary $\mathcal{A}$ and Π fixed above. From each clear training observation it decodes $\theta_u = \mathrm{cl}^{-1}(Y)$ and adds the equation $\Theta_e = \theta_e$ or $q^\top \Psi = \theta_q$, together with the public vertex-parity equations in Equation (15). It solves the graph and compositional systems by fixed Gaussian-elimination rules, setting free variables to zero, and may ignore missing-cue training observations. On a test input with $z \in [K]$, it returns z; on $(u, \perp)$, it returns the class encoded by its reconstructed vector for u. It uses arbitrary fixed outputs on invalid inputs or inconsistent systems, which never occur under the family. Thus $\mathcal{A}_{\mathrm{lin}}$ is a deterministic learner defined on every finite sample, including the empty sample, and always classifies clear observations correctly.

Recovery bounds. Let

$$\zeta := \frac{1}{2}\log_2(4/3) > \frac{1}{5}.$$

We show that there is a finite constant C_λ, depending only on λ, such that, for all sufficiently large n, uniformly over all $v \ge 0$ and all $P \in \mathfrak{P}_{\lambda,n,v}$,

$$\mathbb{E}_S[R_P(\mathcal{A}_{\mathrm{lin}}(S)) - R_P^\star] \le C_\lambda \frac{n}{n+v}\, n^{-\zeta}, \tag{28}$$

$$\mathbb{E}_S\left[R_P\big(\mathcal{A}_{\mathrm{lin}}(S_{I_m^\star})\big) - R_P^\star\right] \le C_\lambda \frac{n}{n+v}\, n^{-1/5}. \tag{29}$$

Missing graph components. Call an edge observed if it occurs in at least one clear graph observation in S. A specified edge has probability $p = a/N$ on each training draw, so $np \ge 1$ and

$$\mathbb{P}\{e \text{ is missing}\} = (1-p)^n \le e^{-1} < \frac{3}{8}.$$

Conditional on a root edge being missing, any s specified other edges are all missing with probability

$$\left(\frac{1-(s+1)p}{1-p}\right)^n = \left(1 - \frac{sp}{1-p}\right)^n \le e^{-s} \le \left(\frac{3}{8}\right)^s. \tag{30}$$

There are at most four first extensions of a path from the root edge and at most two subsequent extensions. If T_e is the connected component of missing edges containing e, a path union bound gives

$$\mathbb{E}[|T_e| \mid e \text{ is missing}] \le 1 + \sum_{\ell \ge 1} 4 \cdot 2^{\ell-1}(3/8)^\ell = 7. \tag{31}$$

Recovery from high girth. The difference between two solutions of the vertex equations that agree on all observed edge vectors is an even subgraph supported on missing edges. Coordinatewise, every such subgraph decomposes into cycles. Hence a specified edge vector is uniquely determined unless the edge lies on an entirely missing cycle. There are at most 2^{k-1} simple cycles of length k containing a fixed edge, and any k specified edges are all missing with probability $(1 - ka/N)^n \le e^{-k} \le (3/8)^k$. Since every cycle has length at least L,

$$\mathbb{P}\{e \text{ lies on an entirely missing cycle}\} \le \sum_{k=L}^{\infty} 2^{k-1}(3/8)^k = 2(3/4)^L. \tag{32}$$

Local oracle certificates. Set

$$s_\star := \lceil n^{1/5} \rceil. \tag{33}$$

For a target edge e, retain one clear example from e if it is observed. If it is missing, consider its entire missing component T_e. Declare certificate failure if $|T_e| > s_\star$ or if e is not a bridge of T_e; otherwise, retain one clear example from every observed edge incident to T_e.

A connected missing component with s edges has at most $s+1$ vertices and therefore at most $3(s+1)-2s = s+3$ incident observed edges. If e is a bridge, remove it and sum the vertex-parity equations over the vertices on either side. Internal edges cancel, and e is the only missing edge crossing the resulting cut. The retained boundary vectors therefore determine Θ_e.

Let C_e be the certificate cost, with $C_e = 0$ on failure. By Equation (31),

$$\mathbb{E}C_e \le (1-p_e) + p_e(7+3) \le \frac{35}{8}, \qquad 0 \le C_e \le s_\star + 3, \tag{34}$$

where $p_e \le 3/8$ is the probability that e is missing. A missing edge is not a bridge precisely when it belongs to a missing cycle, so Markov's inequality and Equation (32) give

$$\mathbb{P}\{\text{the certificate for } e \text{ fails}\} \le \frac{7}{s_\star} + 2(3/4)^L. \tag{35}$$

Recovery of the compositional map. Let L_{c} be the number of clear compositional observations and put $t = p_{\mathrm{c}} - \kappa = \lambda\gamma/20$. Since $L_{\mathrm{c}} \sim \mathrm{Bin}(n, p_{\mathrm{c}})$, Chebyshev's inequality gives

$$\mathbb{P}\left\{ L_{\mathrm{c}} < \frac{p_{\mathrm{c}} + \kappa}{2} n \right\} \le \frac{1}{t^2 n}.$$

Conditional on $L_{\mathrm{c}} = \ell$, the descriptors are independent and uniform in $\mathbb{F}_2^D$. For each nonzero $z \in \mathbb{F}_2^D$, all ℓ descriptors are orthogonal to z with probability $2^{-\ell}$. A union bound therefore gives

$$\mathbb{P}\{\mathrm{rank} < D \mid L_{\mathrm{c}} = \ell\} < 2^{D-\ell}.$$

Thus the full sample determines all of Ψ, and a basis of D clear observations suffices, except on an event of probability

$$q_n := \frac{1}{t^2 n} + 2^{-tn/2} = O_\lambda(n^{-1}). \tag{36}$$

The oracle certificates fit the budget. The oracle knows h. It retains a compositional basis and the successful local certificates of the J active graph edges. Overlap can only reduce the number of retained examples. Let

$$F_h := \sum_{j=1}^{J} C_{e(j,h_j)}.$$

By Equation (34),

$$\mathbb{E}F_h \le \frac{35N}{8g} \le \frac{35}{8}\frac{21\gamma}{20}\frac{\lambda}{100} n < \frac{\lambda\gamma}{20} n, \qquad m - D \ge \frac{\lambda\gamma}{10} n - 1. \tag{37}$$

To control overflow despite the dependence among certificates, observe that toggling one edge occupancy changes at most two missing components before the toggle and at most two afterward. A missing root has nonzero cost only when its component has at most $s_\star$ edges. Thus changed nonzero costs can occur only at the toggled edge or among at most $4s_\star$ edges in the affected small components. Since every cost is at most $s_\star + 3$, one occupancy toggle changes F_h by at most $(4s_\star + 1)(s_\star + 3)$. Replacing one training observation toggles at most two occupancies and hence changes F_h by at most $40 s_\star^2$. The Efron–Stein inequality gives

$$\mathrm{Var}(F_h) \le 800 n s_\star^4.$$

For $n \ge 40/(\lambda\gamma)$, the gap between $m - D$ and $\mathbb{E}F_h$ is at least $\lambda\gamma n/40$. Chebyshev's inequality therefore yields

$$\mathbb{P}\{F_h > m - D\} \le \frac{1{,}280{,}000}{\lambda^2\gamma^2}\frac{s_\star^4}{n} =: u_n = O_\lambda(n^{-1/5}). \tag{38}$$

Completing the recovery bounds. All clear test observations are classified correctly. Averaging Equation (32) over the missing-cue graph domains and using Equation (36) for the compositional domains gives

$$\mathbb{E}_S[R_P(\mathcal{A}_{\mathrm{lin}}(S)) - R_P^\star] \le \omega\big[2a(3/4)^L + \kappa q_n\big]. \tag{39}$$

On compositional-rank failure or budget overflow, the oracle uses the empty subset. Otherwise, it uses the preceding certificates and omits any failed graph certificate. This produces an admissible subset and hence upper-bounds the risk of the optimal subset $I_m^\star$. By Equation (35),

$$\mathbb{E}_S\big[R_P\big(\mathcal{A}_{\mathrm{lin}}(S_{I_m^\star})\big) - R_P^\star\big] \le \omega\left[a\left(\frac{7}{s_\star} + 2(3/4)^L\right) + (a+\kappa)(q_n+u_n)\right]. \tag{40}$$

This calculation averages over the active test domain and does not require simultaneous recovery of every active edge.

Finally, Equation (14) implies

$$(3/4)^L \le \frac{4}{3}(3g)^{\log_2(4/3)} n^{-\zeta} = O_\lambda(n^{-\zeta}).$$

Also $s_\star^{-1} \le n^{-1/5}$, $s_\star^4/n \le 16n^{-1/5}$, $q_n = O_\lambda(n^{-1})$, and $\omega = \gamma n/(100g(n+v))$. Substitution into Equations (39) and (40) proves Equations (28) and (29). The inequality $\zeta > 1/5$ follows from $(4/3)^5 > 4$. Multiplying the two bounds by $(n+v)/n$, then taking the suprema over v and $P \in \mathfrak{P}_{\lambda,n,v}$, proves Equations (4) and (5) and completes the proof. □

B PROOF OF THE KL CONTROL PROPOSITION

Proof. Let $P = \mathcal{A}_T(S)$ and $Q = \mathcal{A}_T(S_I)$, and write $\delta = \mathbb{E}_P q - \mathbb{E}_Q q$. If $D_{\mathrm{KL}}(P\|Q) = \infty$, the claim is immediate. Otherwise, the Donsker–Varadhan variational formula and the σ-sub-Gaussian assumption under Q imply, for every $t > 0$,

$$D_{\mathrm{KL}}(P\|Q) \ge t\delta - \log \mathbb{E}_Q \exp\big\{t\big(q - \mathbb{E}_Q q\big)\big\} \ge t\delta - \frac{t^2\sigma^2}{2}.$$

Optimizing over t bounds the positive part of δ by $\sqrt{2\sigma^2 D_{\mathrm{KL}}(P\|Q)}$. Applying the same argument to $-q$ bounds its negative part.

For the sharper first inequality in **??**, apply the preceding argument to the identity query under the pushforward laws $q_\# P$ and $q_\# Q$. The second inequality follows from data processing:

$$D_{\mathrm{KL}}(q_\# P\|q_\# Q) \le D_{\mathrm{KL}}(P\|Q).$$

□

C PROOF OF ??

Theorem C.1. *Let $q : \mathbb{R}^d \to \mathbb{R}$ be a query function that is σ-sub gaussian under $\mathcal{A}_T(I)$. For any randomized learning algorithm, the expected leave-group-out error is bounded by:*

$$|\mathbb{E}_{\theta\sim\mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta'\sim\mathcal{A}_T(I)}[q(\theta')]| \le \sqrt{2\sigma^2 KLPQ} \tag{41}$$

where $P = \mathrm{Law}(\mathcal{A}(S))$ and $Q = \mathrm{Law}(\mathcal{A}(S \setminus x))$

Proof. The proof is presented for a single point x for brevity, but can be extended for any subset of the training set witout loss of generality. Denoting $\mathrm{LOO}_S := \mathbb{E}_{\theta\sim\mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta'\sim\mathcal{A}_T(I)}[q(\theta')]$, and using the Donsker-Varadhan representation of the KL divergence, we have that:

$$D_{\mathrm{KL}}PQ = \sup_{g\in\mathcal{G}} \mathbb{E}_{\theta\sim P}[g(\theta)] - \log \mathbb{E}_{\theta\sim P}[\exp(g(\theta))]$$

Using $f(\theta) = \lambda(R_D(\theta) - \mathbb{E}_{\theta\sim Q}[R_D(\theta)])$, we have:

$$\begin{aligned} D_{\mathrm{KL}}PQ &\geq E_{\theta\sim P}[f(\theta)] - \log \mathbb{E}_{\theta\sim Q}[\exp(f(\theta))] \\ &= \mathrm{LOO}_S(x) - \log \mathbb{E}_{\theta\sim Q}[\exp(f(\theta))] \\ &\geq \mathrm{LOO}_S(x) - \frac{\lambda^2\sigma^2}{2} \qquad \text{(By the sub gaussian assumption)} \end{aligned}$$

Finally, optimizing over λ yields:

$$\mathrm{LOO}_S(x) \leq \sqrt{2\sigma^2 D_{\mathrm{KL}}PQ}$$

a similar analysis for $-\mathrm{LOO}_S(x)$ gives an upper bound in absolute value. □

Theorem C.2 (KL Output-space upper bound on LOO)**.** *Let $S = \{x_1, \ldots, x_n\}$ be a training set and $\mathcal{A}$ a randomized learning algorithm. For any point $x \in S$ and any query function f such that $f_x(\theta)$ is σ-subgaussian when $\theta \sim \mathcal{A}(S)$, we have that:*

$$|\mathrm{LOO}_S(x)| \leq \sqrt{2\sigma^2 D_{\mathrm{KL}} f_x\#P f_x\#Q}.$$

Proof. The proof follows from a similar argument than the previous theorem, and is obtained by setting $g(\theta) = -\lambda f(\theta)$. Note that Theorem C.1 can be derived from this result by using the Data Processing Inequality. □

D PROOF OF ??

Theorem D.1. *Let*

$$F_n(\theta) := \frac{1}{n}\sum_{i=1}^{n} f_i(\theta), \qquad F_I(\theta) := \frac{1}{m}\sum_{i\in I} f_i(\theta), \qquad \Delta_I(\theta) := F_I(\theta) - F_n(\theta),$$

where $I \subset [n]$, $|I| = m$. Fix $h > 0$, $\beta > 0$, and consider the ULA chains

$$\theta_{k+1} = \theta_k - h\nabla F_n(\theta_k) + \sqrt{2h/\beta}\,\xi_{k+1}, \qquad \theta'_{k+1} = \theta'_k - h\nabla F_I(\theta'_k) + \sqrt{2h/\beta}\,\xi'_{k+1},$$

with $\xi_{k+1}, \xi'_{k+1} \overset{\text{i.i.d.}}{\sim} N(0, I_d)$ and common initial law $\mu_0 := \mathrm{Law}(\theta_0) = \mathrm{Law}(\theta'_0)$. Write $\mu_k := \mathrm{Law}(\theta_k)$ and $\mu_k^I := \mathrm{Law}(\theta'_k)$. Then, for every $T \geq 1$,

$$D_{\mathrm{KL}}\big(P^n_{0:T} \,\|\, P^I_{0:T}\big) = \frac{\beta h}{4}\sum_{k=0}^{T-1} \mathbb{E}_{\mu_k}\left[\|\nabla\Delta_I(\theta_k)\|^2\right], \tag{42}$$

$$D_{\mathrm{KL}}\big(\mu_T \,\|\, \mu_T^I\big) \leq \frac{\beta h}{4}\sum_{k=0}^{T-1} \mathbb{E}_{\mu_k}\left[\|\nabla\Delta_I(\theta_k)\|^2\right], \tag{43}$$

where $P^n_{0:T}$ and $P^I_{0:T}$ denote the path laws of $(\theta_0, \ldots, \theta_T)$ and $(\theta'_0, \ldots, \theta'_T)$.

Proof. The one-step kernels $K_n(\theta, \cdot)$ and $K_I(\theta, \cdot)$ are Gaussians with means $\theta - h\nabla F_n(\theta)$ and $\theta - h\nabla F_I(\theta)$, respectively, and common covariance $\frac{2h}{\beta} I_d$. Since the covariances agree, we have

$$D_{\mathrm{KL}}(K_n(\theta, \cdot) \,\|\, K_I(\theta, \cdot)) = \frac{\beta h}{4}\|\nabla\Delta_I(\theta)\|^2.$$

Both path laws share the initial measure μ_0, so by the chain rule for KL on Markov chains, $D_{\mathrm{KL}}(P^n_{0:T}\|P^I_{0:T}) = \sum_{k=0}^{T-1}\mathbb{E}_{\mu_k}[D_{\mathrm{KL}}(K_n(\theta_k, \cdot)\|K_I(\theta_k, \cdot))]$, which gives equation 42. The bound equation **??** follows by data processing under $(\theta_0, \ldots, \theta_T) \mapsto \theta_T$. □

E PROOF OF THE QUADRATIC PROGRAM FORMULATION OF SUBSET SELECTION

Proof. Recall the upper bound provided in Theorem 5.1:

$$D_{\mathrm{KL}}(\mathcal{A}_T(S) \,\|\, \mathcal{A}_T(I)) \leq \frac{\beta h}{4} \sum_{k=0}^{T-1} \mathbb{E}_{\mathcal{A}_k(S)} \left[\|\nabla \Delta_I(\theta_k)\|^2 \right]$$

We focus on the term $D = \sum_{k=0}^{T-1} \mathbb{E}_{\mathcal{A}_k(S)} \left[\|\nabla \Delta_I(\theta_k)\|^2 \right]$: for $I \subseteq S$, let $\alpha \in \{0,1\}^n$ such that $\alpha_i = \mathbb{1}(Z_i \in I)$. Then, we have that:

$$\begin{aligned}
D &= \sum_{k=0}^{T-1} \mathbb{E}_{\mathcal{A}_k(S)} \left[\|\nabla \Delta_I(\theta_k)\|^2 \right] \\
&= \sum_{k=0}^{T-1} \mathbb{E}_{\mathcal{A}_k(S)} \left[\left\| \frac{1}{m} \sum_{i=1}^{n} \alpha_i \nabla f_i(\theta_k) \right\|^2 - \frac{2}{m} \sum_{i=1}^{n} \alpha_i \langle \nabla f_i(\theta_k), \nabla F_n(\theta_k) \rangle + \|\nabla F_n(\theta_k)\|^2 \right] \\
&= \sum_{k=0}^{T-1} \mathbb{E}_{\mathcal{A}_k(S)} \left[\frac{1}{m^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j \langle \nabla f_i(\theta_k), \nabla f_j(\theta_k) \rangle - \frac{2}{m} \sum_{i=1}^{n} \alpha_i \langle \nabla f_i(\theta_k), \nabla F_n(\theta_k) \rangle + \|\nabla F_n(\theta_k)\|^2 \right]
\end{aligned}$$

□

F PROOF OF ??

Since P_0 is a Gram matrix, it can be factored as $(P_0)_{ij} = \langle \phi_i, \phi_j \rangle$. With

$$\bar{\phi} := \frac{1}{n} \sum_{i=1}^{n} \phi_i, \qquad z_i := \phi_i - \bar{\phi}, \qquad (P_0)_{ij} := \langle z_i, z_j \rangle,$$

we have $\sum_i z_i = 0$ and, on the Boolean cardinality slice

$$\mathcal{B}_m = \left\{ \alpha \in \{0,1\}^n : \mathbf{1}^\top \alpha = m \right\}.$$

the discrepancy is

$$D(\alpha) := \left\| \frac{1}{m} \sum_{i=1}^{n} \alpha_i \phi_i - \bar{\phi} \right\|^2 = \frac{1}{m^2} \alpha^\top P_{\mathrm{c}} \alpha = f(\alpha) + \|\bar{\phi}\|^2. \tag{44}$$

Thus D and the objective of **??** have identical excesses and exchange differences. For the remainder of this subsection, we write $P := P_{\mathrm{c}}$, so

$$D(\alpha) = \frac{1}{m^2} \alpha^\top P \alpha, \qquad h(\alpha) := \nabla D(\alpha) = \frac{2}{m^2} P \alpha, \qquad \succeq 0.$$

Equivalently, one may retain an uncentered representation $f(\alpha) = m^{-2} \alpha^\top P_0 \alpha + b^\top \alpha + c$ and use $h(\alpha) = 2P_0 \alpha / m^2 + b$; all gradient-based identities below are unchanged. Fix

$$\alpha^\star \in \operatorname*{arg\,min}_{\alpha \in \mathcal{A}_m} D(\alpha), \qquad D^\star := D(\alpha^\star), \qquad H_t := D(\alpha_t) - D^\star, \qquad r := \min\{m, n-m\},$$

and write $[x]_+ := \max\{x, 0\}$.

Suppose $h_t := h(\alpha_t)$ is stored. The pair

$$i_t \in \operatorname*{arg\,max}_{i : \alpha_{t,i} = 1} h_{t,i}, \qquad j_t \in \operatorname*{arg\,min}_{j : \alpha_{t,j} = 0} h_{t,j}$$

minimizes the linearized change $h_{t,j} - h_{t,i}$. Define $\gamma_t := h_{t,i_t} - h_{t,j_t}$ and $\delta_t := e_{j_t} - e_{i_t}$.

Algorithm 1 Gradient-ranked single-swap descent

Draw $\alpha_0 \sim \mathrm{Unif}(\mathcal{B}_m)$ and query $P\alpha_0$ to store h_0.
Compute $D(\alpha_0)$ from $P\alpha_0$ and set $\underline{D}_{-1} := -\infty$.
for $t = 0, 1, \ldots$ **do**
    Choose i_t, j_t, γ_t, and δ_t as above.
    Set $\underline{D}_t := \max\{\underline{D}_{t-1}, D(\alpha_t) - r[\gamma_t]_+\}$ and $\widehat{H}_t := D(\alpha_t) - \underline{D}_t$.
    if $\gamma_t \le 0$ **then**
        return $(\alpha_t, \widehat{H}_t)$.
    Set $\widetilde{\alpha}_t := \alpha_t + \delta_t$, query $P\widetilde{\alpha}_t$, and store $\widetilde{h}_t := h(\widetilde{\alpha}_t)$.
    Compute $\Delta_t := \frac{1}{2}\delta_t^\top(h_t + \widetilde{h}_t)$.
    if $\Delta_t < 0$ **then**
        Set $(\alpha_{t+1}, h_{t+1}) := (\widetilde{\alpha}_t, \widetilde{h}_t)$ and store $D(\alpha_{t+1}) := D(\alpha_t) + \Delta_t$.
    else
        return $(\alpha_t, \widehat{H}_t)$.

Lemma F.1 (Exact change and gradient-gap certificate)**.** *Whenever the candidate is tested, its exact objective change is*

$$\Delta_t = -\gamma_t + \frac{1}{m^2}\delta_t^\top P\delta_t. \tag{45}$$

Moreover, at every iterate,

$$H_t = D(\alpha_t) - D^\star \le r[\gamma_t]_+. \tag{46}$$

In particular, the right-hand side is a computable certificate of suboptimality, and $\gamma_t \le 0$ implies that α_t is globally optimal.

Proof. For any feasible $\alpha, \alpha + \delta$,

$$D(\alpha+\delta) - D(\alpha) = \frac{1}{2}\delta^\top\big(h(\alpha) + h(\alpha+\delta)\big) = \delta^\top h(\alpha) + \frac{1}{m^2}\delta^\top P\delta.$$

Since $\delta_t^\top h_t = -\gamma_t$, this proves equation 45.

Fix an optimum $\alpha^\star$. Relative to $\alpha^\star$, the iterate α_t contains $k \le r$ extra indices $i_1, \ldots, i_k$ and misses k indices $j_1, \ldots, j_k$. By convexity,

$$H_t \le h_t^\top(\alpha_t - \alpha^\star) = \sum_{\ell=1}^{k}(h_{t,i_\ell} - h_{t,j_\ell}) \le k\gamma_t \le r\gamma_t.$$

If $k = 0$, then $H_t = 0$; otherwise the last display and $H_t \ge 0$ imply $\gamma_t \ge 0$. This proves equation 46. □

Corollary F.2 (Monotone certified excess)**.** *Every iterate supplies the lower bound*

$$D^\star \ge D(\alpha_t) - r[\gamma_t]_+.$$

Define the best lower bound observed through iteration t and its associated certified excess by

$$\underline{D}_t := \max_{0\le s\le t}\{D(\alpha_s) - r[\gamma_s]_+\}, \qquad \widehat{H}_t := D(\alpha_t) - \underline{D}_t. \tag{47}$$

Then

$$H_t \le \widehat{H}_t, \qquad \widehat{H}_{t+1} \le \widehat{H}_t \tag{48}$$

whenever iteration $t+1$ is reached. Equivalently,

$$H_t \le \widehat{H}_t = \min_{0\le s\le t}\big\{r[\gamma_s]_+ - \big(D(\alpha_s) - D(\alpha_t)\big)\big\} \le r\min_{0\le s\le t}[\gamma_s]_+. \tag{49}$$

Proof. The lower bound follows from Lemma F.1; hence $\underline{D}_t \le D^\star \le D(\alpha_t)$, which proves $0 \le H_t \le \widehat{H}_t$. If iteration $t+1$ is reached, then $D(\alpha_{t+1}) < D(\alpha_t)$, whereas $\underline{D}_{t+1} \ge \underline{D}_t$, proving monotonicity. Finally, substituting the definition of $\underline{D}_t$ gives the equality in equation 49; its last inequality uses $D(\alpha_s) \ge D(\alpha_t)$ for every $s \le t$. □

Theorem F.3 (Suboptimality of the scalable swap)**.** *Let T be the final iteration of* **??**, *let $\alpha_{\text{out}} := \alpha_T$, and let γ_{out} and δ_{out} denote the gradient gap and exchange direction at that iteration. The algorithm returns the valid certificate $\widehat{H}_T$, with*

$$D(\alpha_{\text{out}}) - D^\star = f(\alpha_{\text{out}}) - \min_{\alpha \in \mathcal{A}_m} f(\alpha) \le \widehat{H}_T \le r \min_{0 \le s \le T} [\gamma_s]_+ \le r[\gamma_{\text{out}}]_+. \tag{50}$$

More precisely, if $\gamma_{\text{out}} \le 0$, then $D(\alpha_{\text{out}}) = D^\star$. Otherwise, the tested candidate fails the exact check and

$$0 \le D(\alpha_{\text{out}}) - D^\star \le r\gamma_{\text{out}} \le \frac{r}{m^2} \delta_{\text{out}}^\top P \delta_{\text{out}}. \tag{51}$$

Thus $\widehat{H}_T$ is directly observable and is never weaker than the last-iterate certificate $r[\gamma_{\text{out}}]_+$.

Proof. Every accepted swap strictly decreases D, so the algorithm terminates because $\mathcal{B}_m$ is finite. The bound equation **??** follows from Corollary F.2. If the final candidate is rejected, then $\Delta_{\text{out}} \ge 0$, and equation 45 gives $\gamma_{\text{out}} \le \delta_{\text{out}}^\top P \delta_{\text{out}} / m^2$, proving equation **??**. □

Remark F.4 (Why history helps)**.** *The gaps γ_t need not decrease: an exchange changes both the gradient and the selected and unselected index sets. In contrast, $\widehat{H}_t$ decreases monotonically. Since the choice $s = t$ appears in equation 49, $\widehat{H}_t \le r[\gamma_t]_+$, so the history-aware certificate is always at least as tight as the certificate from the last iterate. It can also be strictly tighter than $r \min_{s \le t} [\gamma_s]_+$, because it subtracts the objective improvement $D(\alpha_s) - D(\alpha_t)$ made after an earlier certificate was computed. Maintaining it requires only two additional scalars and no additional P-matvecs.*

Remark F.5 (Cost and limitation)**.** *The initial gradient costs one P-matvec, and each attempted swap costs one additional P-matvec. The indices i_t and j_t are found by two linear scans, so the remaining work is $O(n)$ per attempt and the working memory is $O(n)$. The method neither forms P, accesses $P_{i_t j_t}$, nor enumerates the $m(n-m)$ feasible pairs. Failure of the tested candidate does not imply one-swap local optimality: another pair may have a smaller quadratic curvature and still decrease D. Tracking $\underline{D}_t$ and $\widehat{H}_t$ adds only $O(1)$ arithmetic and memory per iteration.*

G RANDOM SELECTION GAP

Let U denote the uniform distribution over $\mathcal{B}_m = \{\alpha \in \{0,1\}^n : \mathbf{1}_n^\top \alpha = m\}$. We compute the expected value of the quadratic program objective $f(\alpha) = \frac{1}{m^2} \alpha^\top P \alpha + \alpha^\top q$ and the centered discrepancy $D(\alpha) = f(\alpha) + \frac{1}{n^2} \mathbf{1}_n^\top P \mathbf{1}_n$ under U. By linearity of expectation,

$$\mathbb{E}_{\alpha \sim U}[f(\alpha)] = \frac{1}{m^2} \sum_{i=1}^n \mathbb{E}_U[\alpha_i^2] P_{ii} + \frac{1}{m^2} \sum_{i \ne j} \mathbb{E}_U[\alpha_i \alpha_j] P_{ij} + \frac{m}{n} \mathbf{1}_n^\top q.$$

Using $\mathbb{E}_U[\alpha_i^2] = \frac{m}{n}$, $\mathbb{E}_U[\alpha_i \alpha_j] = \frac{m(m-1)}{n(n-1)}$ for $i \ne j$, and $q = -\frac{2}{mn} P \mathbf{1}_n$, we obtain

$$\begin{aligned}
\mathbb{E}_{\alpha \sim U}[f(\alpha)] &= \frac{1}{m^2} \left[\frac{m}{n} \operatorname{tr}(P) + \frac{m(m-1)}{n(n-1)} \left(\mathbf{1}_n^\top P \mathbf{1}_n - \operatorname{tr}(P) \right) \right] - \frac{2}{n^2} \mathbf{1}_n^\top P \mathbf{1}_n \\
&= \frac{n-m}{mn(n-1)} \operatorname{tr}(P) + \left[\frac{m-1}{mn(n-1)} - \frac{2}{n^2} \right] \mathbf{1}_n^\top P \mathbf{1}_n \\
&= \frac{n-m}{mn(n-1)} \operatorname{tr}(P) - \frac{n-m}{mn^2(n-1)} \mathbf{1}_n^\top P \mathbf{1}_n + f\left(\frac{m}{n} \mathbf{1}_n \right).
\end{aligned}$$

Since $D(\alpha) = f(\alpha) - f\left(\frac{m}{n} \mathbf{1}_n\right)$, the expected squared gradient discrepancy under uniform random selection without replacement is

$$\mathbb{E}_{\alpha \sim U}[D(\alpha)] = \frac{n-m}{mn^2(n-1)} \left(n \operatorname{tr}(P) - \mathbf{1}_n^\top P \mathbf{1}_n \right).$$

H EMPIRICAL CORRELATION BETWEEN FAITHFULNESS ERROR AND TRAJECTORY GRADIENT DISCREPANCY

Table 1: Mean test accuracy over seeds (all methods) for cifar10 / resnet18.

Method	K	Budget	Accuracy (%)	Time (s)	D
el2n	–	4500	28.70	1724.1	–
facility_location	–	4500	65.65	15.4	–
graphcut	–	4500	70.95	15.0	–
herding	–	4500	69.70	5.7	–
kl_faithful_full_resnet18_sgd	1	4500	69.57	723.5	–
kl_faithful_full_resnet18_sgd	4	4500	69.77	807.6	–
kl_faithful_full_resnet18_sgd	10	4500	69.81	827.9	–
kl_faithful_full_resnet18_sgd	15	4500	69.06	1551.5	–
kl_faithful_full_resnet18_sgd	20	4500	70.13	794.5	–
moderate	–	4500	**72.11**	5.4	–
moso	–	4500	68.02	822.6	–
random	–	4500	69.34	0.0	–
el2n	–	9000	49.29	1687.8	–
facility_location	–	9000	77.94	14.6	–
graphcut	–	9000	78.59	23.4	–
herding	–	9000	80.19	6.1	–
kl_faithful_full_resnet18_sgd	1	9000	80.12	822.5	–
kl_faithful_full_resnet18_sgd	4	9000	80.37	1042.2	–
kl_faithful_full_resnet18_sgd	10	9000	80.13	1498.1	–
kl_faithful_full_resnet18_sgd	15	9000	80.86	1985.0	–
moderate	–	9000	**81.20**	5.3	–
moso	–	9000	80.90	1162.2	–
random	–	9000	80.44	0.0	–
el2n	–	18000	**88.29**	2894.8	–
facility_location	–	18000	87.59	14.6	–
graphcut	–	18000	85.90	63.4	–
herding	–	18000	87.25	8.0	–
kl_faithful_full_resnet18_sgd	1	18000	87.02	919.2	–
kl_faithful_full_resnet18_sgd	4	18000	87.05	1347.8	–
kl_faithful_full_resnet18_sgd	10	18000	87.11	1787.6	–
moderate	–	18000	86.87	4.5	–
moso	–	18000	87.23	861.5	–
random	–	18000	87.10	0.0	–
el2n	–	22500	**91.24**	1113.7	–
facility_location	–	22500	89.66	19.1	–
graphcut	–	22500	88.47	78.6	–
herding	–	22500	88.90	8.0	–
kl_faithful_full_resnet18_sgd	1	22500	88.89	1754.6	–
kl_faithful_full_resnet18_sgd	4	22500	88.91	1407.7	–
kl_faithful_full_resnet18_sgd	10	22500	88.69	1827.6	–
moderate	–	22500	88.25	4.1	–
moso	–	22500	88.92	1135.3	–
random	–	22500	88.67	0.0	–